

Visualizing Data

Your Tasks (Mark these off as you go)

- ☐ Tell a data story
- ☐ Obtain a dataset of objects
- ☐ Identify categorical variables
- ☐ Tidy the data into a computer readable format
- ☐ Create visualizations for your data
- ☐ Analyze your data
- ☐ Receive credit for this lab guide

☐ Tell a data story

With a partner, look through the graphs on this site:

<https://www.nytimes.com/2023/07/26/learning/over-75-new-york-times-graphs-for-students-to-analyze.html>

Scroll a bit and you will notice that the graphs are organized by topic and type,

Graphs Organized by Topic

Environment, Science and Technology

- [Houston May Get 50 Inches of Rain. How Long Does it Take Your City to Get That Much?](#)
- [It's Not Your Imagination. Summers Are Getting Hotter.](#)
- [The Places in the U.S. Where Disaster Strikes Again and Again](#)

Graphs Organized by Type

[Most?](#)

Distribution (values and their frequency)

- [Six Myths About Choosing a Major](#) (boxplot)
- [It's Not Your Imagination. Summers Are Getting Hotter.](#) (histogram)

Paste a graph that you and your partner find interesting below.

What do you notice? Share what you are noticing and what this may imply.

What do you wonder? Where could you find the answers to what you wonder?
What's going on in this graph? Use what you notice. What can you infer from this graph beyond what it shows directly? What's the deeper story that comes from this graph?

□ Obtain a dataset of objects

Locate the datasets of objects you have been provided. Record the data set your group selected.

□ Identify categorical variables

Observe the dataset of objects you selected. What do you notice? Do the objects have different colors, shapes, size, design, etc.
What do you wonder? What you wonder about will depend on the dataset of objects you selected. For example, how many are there of each color? How many are there of each shape? Etc. Write what you wonder about as questions below. You must write two questions about your data set.
Question 1:
Question 2:

A **categorical variable** (also called **qualitative variable**) is a variable that can take on a fixed number of possible values based on some observed quality. List the categorical variables you identified below.

Categorical variable 1:

Categorical variable 2:

□ Prepare the data

It is often said that 80% of data analysis is spent on the process of cleaning and preparing the data. Below is an example of scrunchies organized by color and frequency of wear.

Physical visualization of the dataset	Numerical visualization of the data set												
	WHICH SCRUNCHIES DO YOU USE? <table><thead><tr><th><u>YES, often</u> For school</th><th><u>SOMETIMES</u> For YOGA OR HAVE SOME DAY</th><th><u>MAYBE</u> DURING AND THE AFTER</th><th><u>RARELY</u> ONLY AT HOME</th></tr></thead><tbody><tr><td></td><td></td><td>Black 3 WHITE 2 GRAY 1 YELLOW 2 D OLIVE 2 D GREEN 2 L GREEN 2 L BLUE 1 Pink 3</td><td>OTHER 2 D GREEN 1 Pink 3 D BLUE 1 BROWN 2</td></tr><tr><td>WHITE 1 PINK 1 YELLOW 1 ORANGE 1</td><td>L BLUE 1 GRAY 3 PURPLE 6</td><td></td><td></td></tr></tbody></table> TOTAL 4 10 20 9	<u>YES, often</u> For school	<u>SOMETIMES</u> For YOGA OR HAVE SOME DAY	<u>MAYBE</u> DURING AND THE AFTER	<u>RARELY</u> ONLY AT HOME			Black 3 WHITE 2 GRAY 1 YELLOW 2 D OLIVE 2 D GREEN 2 L GREEN 2 L BLUE 1 Pink 3	OTHER 2 D GREEN 1 Pink 3 D BLUE 1 BROWN 2	WHITE 1 PINK 1 YELLOW 1 ORANGE 1	L BLUE 1 GRAY 3 PURPLE 6		
<u>YES, often</u> For school	<u>SOMETIMES</u> For YOGA OR HAVE SOME DAY	<u>MAYBE</u> DURING AND THE AFTER	<u>RARELY</u> ONLY AT HOME										
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WHITE 1 PINK 1 YELLOW 1 ORANGE 1	L BLUE 1 GRAY 3 PURPLE 6												

Consider the categorical variables you identified above. Organize your objects based on the first categorical variable you identified. Take a picture of your physical visualization and insert this below. Do the same for your second categorical variable.

Refer to your pictures, then create a numerical visualization by counting and recording the number of items in each category. Do this for both of your categorical variables.

□ Tidy your data into a computer readable format

Tidy Data has become the standard format for science and business because it allows people to easily turn a data table into graphs, useful for analysis and insight.

There are three interrelated rules that make a dataset tidy,

1. Each variable is a column; each column is a variable.
2. Each observation is a row; each row is an observation.
3. Each value is a cell; each cell is a single value.

Below is an example of initial data collected in spreadsheet, and entered in a typical way that most novices would use (left), and the same data in Tidy Format (right),

Day	Plant A height (cm)	Plant B height (cm)
1	0.7	1.5
2	1.0	0.7
3	1.5	0.9
4	1.8	1.3
5	2.2	1.8



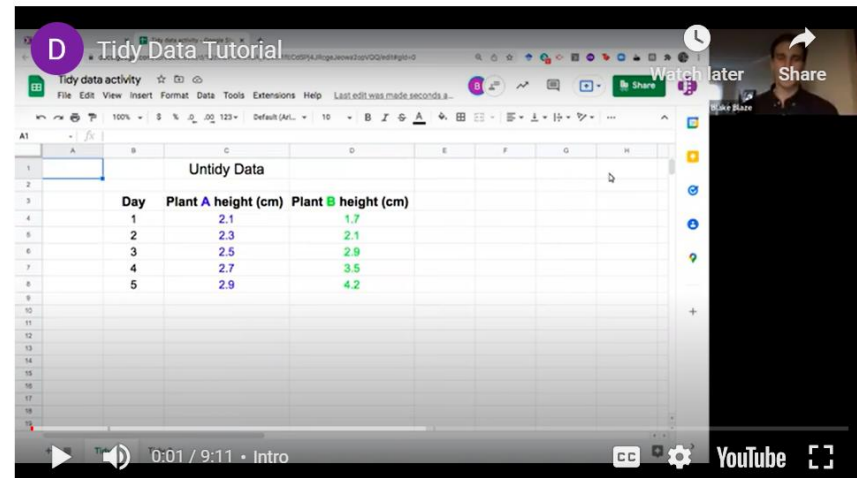
Day	Plant	Height (cm)
1	A	0.7
2	A	1.0
3	A	1.5
4	A	1.8
5	A	2.2
1	B	1.5
2	B	0.7
3	B	0.9
4	B	1.3
5	B	1.8

This is clearly the same data, but it now has:

- Three clearly identified **variables** (days, plant type, and height) each of which has its **own column**.
- Each **row** is an independent **observation** of the variables
- Each **cell** contains a single **value** of a single variable

Watch the video below, to learn more,

Tidy Data Tutorial



The screenshot shows a Google Sheets spreadsheet titled 'Untidy Data'. The data is organized as follows:

Day	Plant A height (cm)	Plant B height (cm)
1	2.1	1.7
2	2.3	2.1
3	2.5	2.9
4	2.7	3.5
5	2.9	4.2

The video player interface includes a play button, a progress bar at 0:01 / 9:11, and a 'Watch later' button. The YouTube logo is visible in the bottom right corner.

<https://youtu.be/fFoK8Rdo4SA?si=GFELBp2hCL8X55zc>

Below is an example of how the scrunchy dataset above was organized in a Tidy format using the categorical variables color and frequency worn.

Scrunchy data set in Tidy format

	A	B
1	Color	Frequency Worn
2	pink	always
3	pink	occasionally
4	pink	occasionally
5	pink	occasionally
6	pink	rarely
7	pink	rarely
8	pink	rarely
9	purple	sometimes
10	purple	sometimes
11	purple	sometimes
12	purple	sometimes
13	purple	sometimes
14	purple	sometimes
15	gray	occasionally
16	gray	occasionally
17	gray	occasionally
18	gray	sometimes
19	gray	sometimes
20	gray	sometimes

Create a new Google Sheet. Enter your data in a Tidy format as explained in the video and illustrated above. Before creating your graphs, check your dataset for missing values, accidental duplicate entries, inconsistent spelling or capitalization, and categories that mean the same thing. Record one change you made—or explain why no change was needed. Share your sheet and include a link to it below.

Google Sheet link:

Data-cleaning change (or why none was needed):

□ Create visualizations for your data

Once you have your data in a Tidy format, it is easy to visualize your data using Google Sheets. Below is an example of how we can visualize the number each color in the scrunchy dataset,

How to visualize the categorical variable, color, in Google sheets

Untitled spreadsheet

File Edit View Insert Format Data Tools Extensions

Color

Color	Frequency
pink	alwa
pink	occa
pink	occa
pink	occa
pink	rare
pink	rare
pink	som

Chart editor

Setup Customize

Chart type

Pie chart

SUGGESTED

Count of Color

Count of Color

Count of Color

A1:A16

A1:A16

A1:A16

You can change the chart type using the drop down menu

Create two visualizations using the categorical variables you identified above. Choose chart types that make comparisons between categories clear. For each visualization, explain why the chart type is appropriate for the data. Take a screenshot of each visualization and paste it below.

□ Analyze your data

Now that you have created your visualizations, revisit your original questions. For each question, state your answer and cite a specific value or pattern from your visualization as evidence.

Question	Answer and Evidence

□ Receive Credit for this lab guide

Submit this portion of the lab to Google Classroom to receive credit for the lab guide.